Younsoo Park

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Education

Pennsylvania State University, University Park, PA — Aug 2021 – Present
B.S. in Computer Science (Minor in Mathematics), Dean's List — Expected Graduation: May 2027

Experience

Levit Inc. (Shopport AI) - Associate Problem Solver — June 2026 – Aug 2026
- Worked with the **CEO** on routing, evaluation, and personalization for Shopport, Levit's live iOS/Android AI app.
- Rebuilt LLM routing as a semantic router (∼**8**× **faster**) + hybrid classifier: **86%** V3 accuracy, **93%** errors rescued.
- Shipped Wowness, a Mem0-backed recommender, as sole engineer; built the evaluation & backfill platform.
- Fixed production failures and analytics false positives; contributed **730 commits** and **143 merged PRs** (+114K LOC).

Hangukgwan - Software Engineering Intern, Zhubei, Taiwan — Summer 2022, Summer 2025
- Built a live QR ordering & POS system for 2 stores (Node.js, MongoDB, Pusher) with ESC/POS kitchen printing.
- Built owner/staff tools: time-card OCR payroll, a 20K-row ingredient ledger (21-month price trends), and P&L.
- Added Claude receipt OCR, ECPay/split payments, a 4-language Next.js site, an Android kiosk app; 200+ test files.

Atom Tech Solutions LTD - Software Engineering Intern, Berkeley, CA — May 2022 – Aug 2022
- Built SQL/JWT authentication, encrypted credential storage, and automated account onboarding.

Research

Federated TinyML for IoT Security - Team Lead, with Dr. Suman Saha & Dr. Peilong Li — Oct 2025 – Present
- Federated, adversarially robust IoT-security TinyML (for LCTES '26): recall 46.7%→93.9%, F1 84.1%→89.3%.
- Compressed models 12.3× and cut ESP32 latency 74.5% across 48 configs (pruning, distillation, INT8 quantization).

Diffusion-Residual Tube MPC for Autonomous Driving — Oct 2026 – Present
- Built an Argoverse 2 pipeline (87K vehicle tracks) and physics baselines for diffusion-residual trajectory prediction.
- Showed baseline error is anisotropic and maneuver-dependent (6 s FDE p95: 31 m), motivating scene-aware margins.

AI Agent Memory Systems - with Prof. Kiwan Maeng — Jul 2026 – Present
- Researched 5 agent memory frameworks (Mem0, A-MEM, LangMem, Zep, OpenAI Memory) to identify the SOTA.

Leadership

Advanced Vehicle Team (EcoCAR) - Project Manager, Vehicle Systems Engineering — Sept 2026 – Present
- Managed Smartsheet schedules for 4 vehicle-engineering deliverables: owners, milestones, dependencies, deadlines.
- Coordinated software, hardware, controls, safety, and testing teams on approvals, facility and vehicle access.
- Built Year 1 strategic drivers, critical-path analysis, and risk plan for EcoCAR (Argonne, GM, MathWorks).

Republic of Korea Air Force - Squad Leader — Sept 2022 – June 2024
- Promoted to Squad Leader and led 20+ airmen in high-tempo operations; received a Commendation Award.

THON™, Pennsylvania State University - Technology Captain — Apr 2026 – Present
- Lead year-round technology operations and ship features for a fundraiser that raises $10M+ annually.

ACM - Web & Design Director — Apr 2026 – Present

Pennsylvania State University - Lead Teaching Assistant, Math 230 (Multivariable Calculus) — Aug 2025 – Present

Projects

OONA (formerly XIIO) – Video & Network Platform for Filmmakers — Apr 2026 – Present
- Building a LinkedIn-style platform for film creators (Next.js, Firebase, Cloudflare Stream, Stripe) plus an Expo app.

SIGNUM (Nittany AI Alliance – PIT-UN Funded) - Co-founder — Aug 2024 – Nov 2025
- Co-founded a $3,750 PIT-UN-funded caregiver AI platform; built a HIPAA-aware RAG assistant (Bedrock, FAISS).
- Built CMS/NPPES/Google Places ETL into DuckDB, cutting caregiver research from 50+ hrs/week to near-instant.

GladIEEEators – IEEE Battle Bots (1st Place) — Apr 2026
- Designed Shot & Chaser, a two-robot combat system with a 334 J AR500 spinner, FEA-validated on a $976 budget.

Skills

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL
Frameworks: React, Next.js, Node.js, NestJS, Express, Flask, Firebase, Tailwind CSS
AI/ML: PyTorch, TensorFlow, Flower, scikit-learn, FAISS, Mem0, AWS Bedrock, OpenAI API, RAG
Data & DevOps: PostgreSQL, MongoDB, Redis, DuckDB, Docker, Kubernetes, Helm, GitHub Actions, Datadog
Project Management: Smartsheet, WBS, Gantt Charts, Critical Path Analysis, Risk Management, SharePoint

Awards & Honors

- 1st Place, ThinkNeuro LLC Project Competition — Apr 2026
- 1st Place, IEEE Student Chapter Battle Bots Competition — Apr 2026
- Pennsylvania State University Scholarships & Grants ($6,500 total) — Fall 2026
- PIT-UN Funding Award ($3,750) – Nittany AI Challenge — Apr 2025
- 3rd Place, ACM Pathfinder Robotics Challenge — Apr 2025